

WU, YU-HSUAN

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EDUCATION

National Yang Ming Chiao Tung University (NYCU)

2028 (expected)

Master of Arts in Linguistics

- Relevant Coursework (expected): Phonology, Computational Linguistics

National Taiwan University (NTU)

2026

Bachelor of Arts in Foreign Languages and Literatures

- Member of NTU Computational Linguistics Specialized Program
- Member of NTU Chinese-English Translation Program
- Relevant Coursework: NLP, Computational Linguistics, Phonology, Semantics

EXPERIENCE

Cathay Financial Holdings — DDT AI

Taipei, Taiwan

ML/AI Engineer Intern

Feb 2026 – Present

- Optimized AI News agent orchestration logic, data ETL, LINE integration and AWS deploy architecture.
- Built a recording-ASR-based BRD auto-revising agent with human-in-the-loop audit.
- Deployed a speech-driven form-filling service using streaming LLM on AWS for enterprise usage.

National Taiwan University

Taipei, Taiwan

Research Assistant — Prof. Shu-hao Shih

Jun 2024 – Jun 2026

- Designed perception experiment schema, prepared stimuli, and analyzed data for non-moraic schwa stress-bearing naturalness.
- Surveyed artificial-learning-experiment related phonology papers and prepared experiment stimuli using Praat and PsychoPy.

Taiwan AI Academy (AIA)

New Taipei City, Taiwan

AI Developer & Technical Support Summer Intern

Jun 2025 – Aug 2025

- Architected a LINE LIFF App career counseling chatbot with an n8n-based RPA workflow and managed user data with Supabase.
- Supported Enterprise Cloud Platform education courses as technical support.

PROJECTS

AI News Agent — Three-Scenario Routing Chatbot

Cathay Financial Holdings, DDT AI

- Built an internal AI research report query assistant using a four-layer architecture (Interface / Application / Data / External Services).
- Designed a three-scenario classification system (LLM intent detection + Python state machine) for intelligent routing across explicit, fuzzy, and out-of-scope queries.
- Implemented asynchronous Lambda self-invocation to overcome LINE Webhook's 1-second timeout, supporting 5–15 second LLM inference + RAG retrieval pipelines.
- Built a full ETL pipeline (Google Sheets → Bedrock Embedding → OpenSearch) with multi-dimensional filtering by topic, source, date, and subscription tier.

BU Agent — Web-Based Two-Mode BRD Co-Drafting Assistant

Cathay Financial Holdings, DDT AI

- Built an internal web app using LangGraph + PostgresSaver with a split layout (chat ↔ mode-aware workspace); BU Agent serves as the BU's single entry point (two modes) to draft v1.0 BRD, then hands off structured deliverables to the downstream BA Agent.
- Explore mode: 5-stage question bank (diverge → converge → challenge) with a live 5-dimension rubric workspace; emits a dual handoff (JSON + 200–400-word paragraph) that auto-transitions to Consult, compressing ideation to 5–10 min.
- Consult mode: Step 1 auto-drafts the BRD outline from per-BU YAML templates and adding status tags for each sections; Step 2 interviews only “needs_round2” tag with live-preview inline edit, keeping BU interview time to 20–30 min.

Voice-Notion — Cloud-First Standup Assistant

Cathay Financial Holdings, DDT AI

- Built a Slack bot that converts mp4 work oral report into confirmable Notion pages and tasks, plus a daily automated standup summary
- Re-architected a single-instance Next.js service into a cloud-native ECS Fargate deployment with DynamoDB, SQS and EventBridge to enable horizontal scaling and zero-data-loss rolling updates
- Split webhook handling so web verified signatures, enqueued to SQS and returned 200 within 3s while worker handled 30–120s transcription and LLM work
- Evaluated Lambda + API Gateway, self-managed EC2 and App Runner against ECS Fargate on persistent worker, cold start/ffmpeg, HA, ops cost and private-subnet criteria

PUBLICATIONS & TALKS

Examining LLM's Ability to Judge Presuppositions: A Comparative Study of Zero-Shot and Self-Discover-NLI Prompting Oct 2025

- Developed and presented the "Self-Discover-NLI" prompt engineering framework, adapting the three-step Self-Discover pipeline (SELECT → ADAPT → APPLY) specifically for Natural Language Inference (NLI) tasks.
- Conducted systematic experiments on the IMPPRES dataset (95 tasks per trigger type), evaluating GPT-4o's handling of clefts, conditionals, modals, and other linguistic triggers.
- Achieved ~10% overall accuracy improvement; performed in-depth error analysis identifying systematic model weaknesses.

SKILLS

Languages: Mandarin (Native), English (Fluent)

Programming: Python

Cloud & Infrastructure: AWS (Lambda, Bedrock, OpenSearch, DynamoDB, Amazon ECR, EventBridge, Amazon SQS)

Frameworks: LangGraph, LangChain

Tools: Microsoft Office, AWS CLI (register-task-definition, create-service, update-service, logs tail), Docker (node:20-slim image with ffmpeg, --no-cache builds)

Certificates: TOEIC (Listening and Reading 985, Speaking 180, Writing 170), AWS SAA (Solutions Architect Associate)